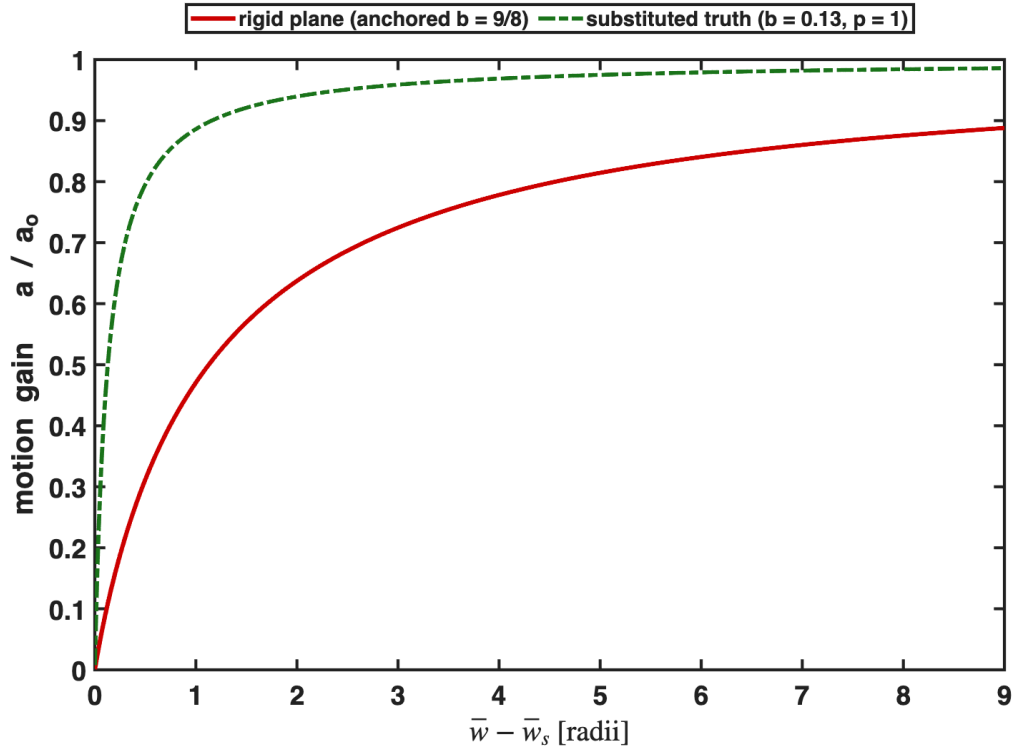


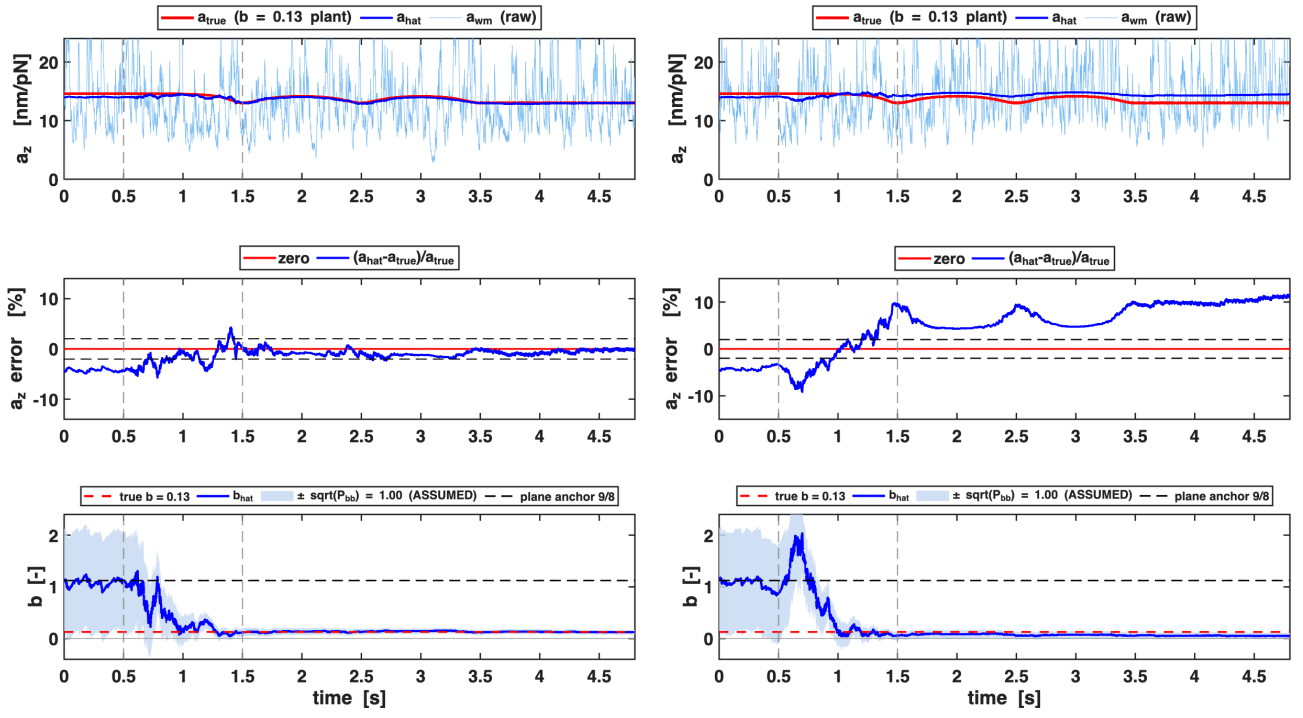


The rigid plane our priors are anchored to (red), and the truth substituted on the next page (dash-dot): same family, same contact point, 63.5 prior widths apart.

The controller does not estimate the gain height by height. It assumes the gain follows one curve,

$$\bar{a} = 1 - \left[\frac{b}{(\bar{w} - \bar{w}_s) + b} \right]^p,$$

and estimates only that curve's constants: $\bar{w}_s$, the contact position, and b , the length over which the wall effect is spent. Whatever those constants converge to, the gain is read off that shape and no other.



Widening the b prior until the truth is inside it — the widths are assumed, not derived: the filter does reach the truth, but the answer becomes seed-dependent.

$Q = 0$ is physically right — b is a constant, it does not drift. It also means P can only shrink: the filter gets one chance, and keeps whatever noise that particular run happened to carry. Raising Q is not the fix either. It would claim b drifts, which is false.

Neither choice is right, so what is missing is not a value of Q .

The model can say *this constant does not change*.

It has no way to say *my belief about it may be wrong*.

Both are carried by the same Q .

The run itself already carries the evidence that the belief is wrong. There is nowhere in the model to put it.